

Curvature and fracturing based on global positioning system data collected at Sheep Mountain anticline, Wyoming

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ABSTRACT

We investigate the curvature-fracture relationship at Sheep Mountain anticline, Wyoming, by coupling fracture mapping with structural mapping using high-precision global positioning system data. Carrier-phase post-processing techniques of spatial data collected across patches of bedding surfaces result in a high-resolution data set. Differential geometry tools form the basis for curvature analysis, allowing for a quantitative understanding of the shapes of these surfaces. Comparison of principal curvature magnitudes with fracture measurements indicates that greater curvature correlates with greater spherical variance of fracture set orientations. Fracture intensities, however, correlate only loosely with curvature, because fracturing mechanisms other than curvature of bedding must be taken into account.

Keywords: fractures, folds, GPS, intensity, variance analysis.

INTRODUCTION

Outcrop-scale brittle structures such as small faults, joints, and sheared joints form within folding sedimentary rock. These structures represent a second order of deformation, developing as localized deformation features within more brittle lithologies and relieving local stress perturbations as the deforming strata bend into various shapes. As small-scale structural heterogeneities, the faults, joints, and sheared joints affect the flow properties of reservoirs and aquifers. They represent dis-

continuities in the permeability of the rock volume that influence both the vertical and lateral transport of fluids. Accurate characterization of these structures within reservoirs or aquifers is sought for economic purposes.

Sedimentary horizons and major faults can be satisfactorily imaged in seismic reflection data (e.g., Fiore et al., 2007; Kattenhorn and Pollard, 2001; Maerten et al., 2002; Needham et al., 1996; Mansfield and Cartwright, 1996). Identifying so-called subseismic fractures (with length scales ≤ 20 –30 m), however, is problematic. The secondary structures on which this paper focuses are in this category, being typically below seismic resolution. Direct observation of the patterns in which they form is unfeasible, except within boreholes, where spatial coverage is limited. New methods for predicting fracture patterns from available subsurface data would play a crucial role in the development of hydrocarbon reservoirs and groundwater aquifers.

Many previous studies have hypothesized a relationship between fracture location, orientation, spatial density, and structural position across a fold (e.g., Woodring et al., 1940; Harris et al., 1960; Stearns, 1968; Narr, 1991; Cooper, 1992). In recent years, curvature analysis quantifying properties of fold geometry has been emphasized as a means of predicting fracture patterns within folds (e.g., Schultz-Ela and Yeh, 1992; Lisle, 1994; Fischer and Wilkerson, 2000; Hennings et al., 2000). Studies have implemented curvature analysis in various ways. Fischer and Wilkerson (2000) related fracture orientation to minimum curvature trajectories; Lisle (1994), Robinson (1997), and Hennings et al. (2000) correlated joint occurrence and density with Gaussian curvature magnitudes. These studies rely on an assortment of curvature calculation methods (e.g., Murray,

1968; Ekman, 1988; Ivanov, 1989; Schultz-Ela and Yeh, 1992; Lisle, 1994; Lisle and Robinson, 1995; Nothard et al., 1996; Stewart and Podolski, 1998; Johnson and Johnson, 2000; Roberts, 2001). Recently, Bergbauer and Pollard (2003) and Pearce et al. (2006) presented a precise method of surface curvature calculation that is derived from differential geometry.

We investigate the relationship between curvature and fracturing at Sheep Mountain anticline, Wyoming, by comparing the magnitudes of the principal curvatures, derived from differential geometry calculations, of small patches of bedding surfaces with the intensities and orientations of fractures measured across these surfaces.

GEOLOGICAL SETTING

Sheep Mountain anticline is located on the northeast flank of the Bighorn Basin, just west of the Bighorn Mountains. It is a basement-cored thrust fault-related fold that formed in response to Laramide tectonics. The fold trends northwest-southeast and is cut by the Bighorn River approximately perpendicular to this trend (Fig. 1). The study area consists of the portion of the anticline that is northwest of the river cut, and includes sedimentary rocks ranging in age from early Carboniferous to Permian (Fig. 1). The oldest rocks are the Mississippian Madison Formation, a massive limestone that has been dolomitized in much of the study area (Pranter et al., 2005; Sonnenfeld, 1996). The Madison Formation is exposed in the canyon where the Bighorn River dissects Sheep Mountain and in the hinge of the anticline where younger layers have been eroded. The Pennsylvanian Amsden Formation is above a karst surface at the top of the Madison Formation and is composed of a basal sandstone unit, a middle silty shale unit,

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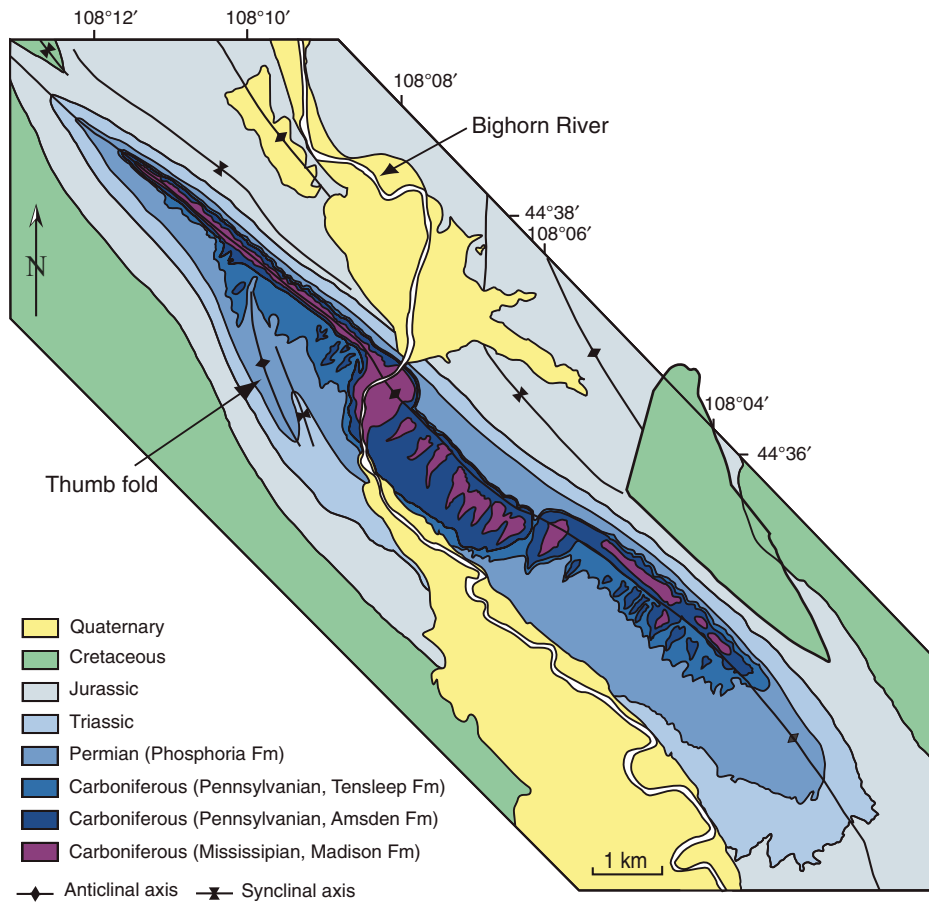


Figure 1. Geological map of Sheep Mountain anticline. The Bighorn River dissects the fold approximately perpendicular to the fold trend. This study focuses on deformation of the Permian Phosphoria limestone (from Bellahsen et al., 2006a; after Rioux, 1994).

and an upper unit of interbedded limestone and dolomite (Ladd, 1979; Hennier, 1984). The Amsden Formation crops out primarily in the hinge of the fold. Above the Amsden Formation, the Pennsylvanian Tensleep Formation consists predominantly of sandstone that is interlayered with thin beds of dolomite and shale. The Tensleep Formation forms large pavements in the backlimb and forelimb of Sheep Mountain. Limited Tensleep outcrops are found in the hinge of the fold near the northwestern nose. The youngest formation in the study area is the Permian Phosphoria Formation, composed of interbedded siltstones and shales overlain by a massive limestone. This limestone forms the flatirons along the steep forelimb of the fold, the folded pavements over the northwest nose, and the small pavements at the base of the backlimb slopes. For this study, we focus on the Phosphoria Formation because it forms fairly continuous pavements both in areas of significant curvature (i.e., the hinge) and apparently planar areas (i.e., base of forelimb and backlimb dipslopes).

METHODOLOGY

Global Positioning System Data Collection

To collect the three-dimensional spatial data analyzed in this study, we used differential global positioning system (GPS) technology with a two receiver setup. A Trimble Pro XRS receiver served as a stationary base station and a Trimble Pro XL receiver with a pole-mounted antenna served as a rover. For two of the pavements considered in this study, we walked across the bedding surfaces with the rover system, but the remaining five pavements were too steep. For these, the rover was kept stationary at distances between 5 and 20 m and offsets to positions on the bedding surfaces were recorded with a LaserCraft Contour XLRic laser range finder.

For all pavements considered in this study, data were collected and post-processed using the carrier-phase (L1) signal, which has a much higher frequency than the more common code

signal (C/A, pseudo random code; Kaplan, 1996). The higher frequency of the carrier signal enables greater precision and accuracy of measurements by orders of magnitude (Kaplan, 1996). The collection of precise GPS data sets is thus feasible on an academic budget.

To determine the effect of various post-processing techniques on collected GPS data, we ran a test case at Jasper Ridge Biological Preserve in Stanford, California. An area of noticeable curvature and of comparable size to pavements intended for analysis in Wyoming was selected at the preserve. Three-dimensional spatial data were collected at approximately regular intervals and then were post-processed by four different methods. To investigate the difference between correcting GPS positions with an on-site base station versus a distant base station, we post-processed the collected positions with data from both a base station we had set up at the preserve and a community base station 28 km away at Pigeon Point, California. To investigate the difference between correcting GPS positions with code data versus carrier-phase data, we post-processed both the code data and the carrier-phase data that we had collected with the roving GPS receiver. The resulting corrected data sets are represented by black dots in Figures 2A–2D. Basic surfaces (MATLAB's triangle based linear interpolation) have been fitted to these points and contoured at a 0.6 m interval for comparison to a digital elevation model (DEM) of the preserve with a resolution of 0.6 m (Fig. 2E). Our results indicate that for post-processing code data, an on-site base station provides more accurate measurements than a base station 28 km away. More notable is the greater accuracy of carrier-phase post-processing. The contours in Figure 2E from a high-resolution DEM are reproduced faithfully in Figures 2C and 2D.

The decision to set up a base station at Sheep Mountain rather than using a community base station was based on the desire to collect large amounts of high-quality data. When post-processing using carrier-phase data, a position solution is generated at the rate of the least common multiple of the base and rover logging intervals. Most community base stations have a 5 s logging interval at minimum. With 5–10 position readings per location required to ensure a measurement of reasonable precision, setting up a base station with a 1 s logging rate allowed us to collect data much more efficiently. In addition, the ability of the post-processing method to synchronize the signals collected by the base station and rover is inversely proportional to the baseline distance. The nearest base station to Sheep Mountain is 180 km away, much beyond the 30 km maximum baseline distance that is

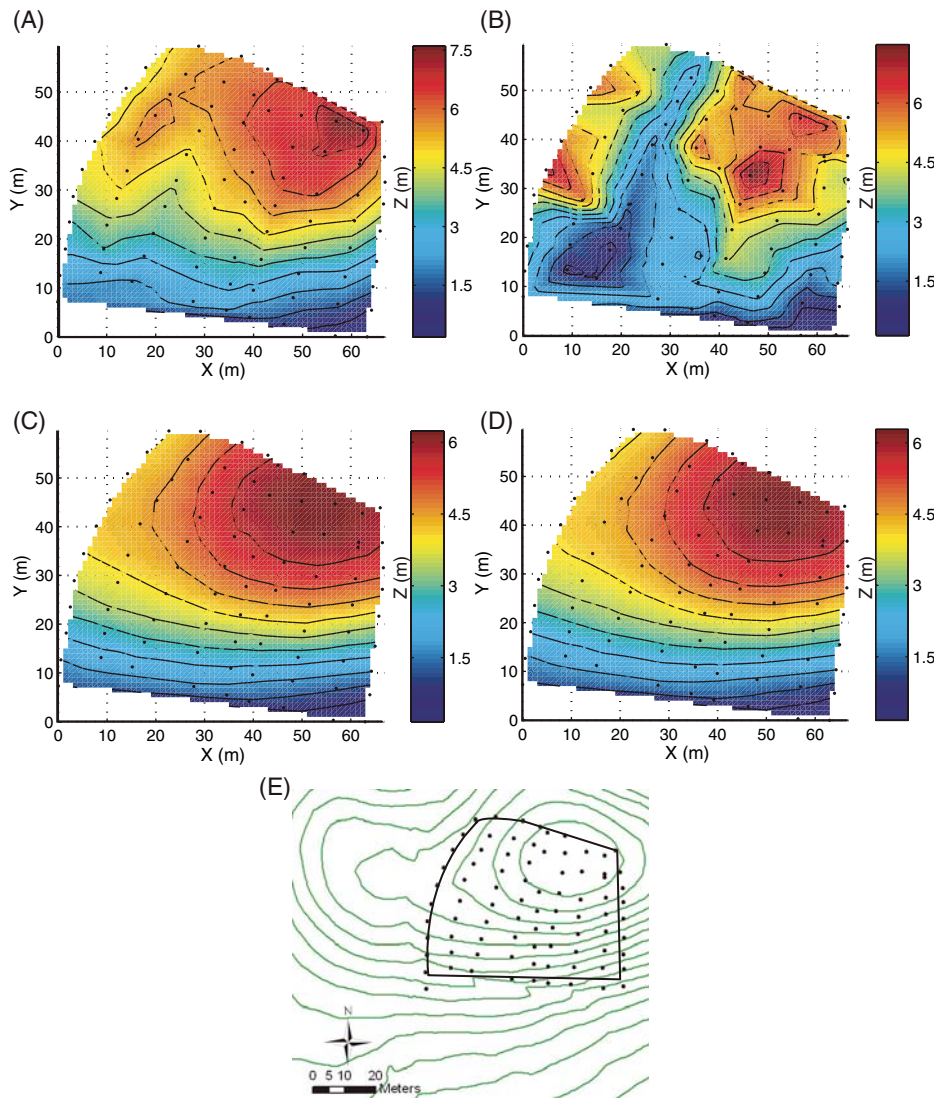


Figure 2. Results of testing various digital global positioning system (GPS) post-processing methods. GPS data are post-processed using (A) code data with an on-site base station; (B) code data with a Trimble referenced base station located 28 km from the field site; (C) carrier-phase data with an on-site base station; (D) carrier-phase data with a Trimble referenced base station located 28 km from the field site. Surfaces have been gridded from the respective post-processed data. Elevation is shown in contours spaced at 0.6 m intervals that can be compared with the digital elevation model (DEM) contours shown in E. (E) Collected data are shown in black. Green lines represent a 0.6 m resolution DEM of the ground surface. The black polygon represents the area that is gridded in A–D.

required for high-precision post-processing (Draffan, 2005, personal commun.).

For this project we are characterizing the shapes of individual patches of bedding surfaces. Therefore, we require high relative accuracy and precision of points within a single surface, but not high global accuracy (positioning of surfaces within a global reference frame). Moving the location of the base station for each characterized surface therefore provides the best

quality data for these purposes. The base station typically remained within 500 m of a characterized surface. The longest baseline from the base station to the rover was 1.8 km.

GPS Data Filtering

To ensure that the appropriate features are analyzed during the curvature calculation, two steps are taken. During the collection

phase, to prevent aliasing effects, the pavement is sampled at a scale that is smaller than the scale of features being studied. During the data processing phase, small-scale undulations that are not related to the phenomenon being considered (i.e., folding) are removed using the spectral analysis technique described by Bergbauer and Pollard (2003). Through this technique, the data are transformed from the spatial to the frequency domain and decomposed into a series of trigonometric functions of varying amplitude, wavelength, and phase (Davis, 1986; Bracewell, 2000). We then specify a maximum frequency threshold so that any data of higher frequency (shorter wavelength) than this threshold value are discarded. This spectral analysis technique provides the opportunity to control the wavelength content of the data set and focus on the scale of folding that is of interest (e.g., Stewart and Wynn, 2000).

Curvature Calculation

The curvature calculation in this study is derived from concepts and equations of differential geometry presented in Bergbauer and Pollard (2003). The shape of nonplanar surfaces can be completely described using the first and second fundamental forms (Struik, 1961). These quantities are related to the arc length of a curve in any direction through a point on a surface and the shape of the surface near that point (Bergbauer and Pollard, 2003; Pollard and Fletcher, 2005). The ratio of the second to the first fundamental form, both of which are invariants with respect to choice of coordinate systems, defines the normal curvature (Struik, 1961). Normal curvature at a point varies with orientation and the extreme values are termed the principal curvatures, κ_1 and κ_2 , respectively, the maximum and minimum curvature values that are associated with orthogonal curves in the surface. Gaussian curvature is the product of these maximum and minimum curvature values.

Fracture Data Collection

Fracture orientation data were collected across each of the selected pavements. Mode of deformation (opening or shearing), evidence for reactivation, and intensities also were recorded. Intensity measurements were obtained along scan lines laid perpendicular to the strike of the fracture set and parallel to bedding. The number of fractures intersecting the scan line was counted and recorded along with the length of the scan line so that intensity measurements could later be normalized to represent the number of fractures per meter. Bed thicknesses were recorded at each site.

Fracture Data Analysis

Fisher analysis is used to determine the dispersion in orientation of fractures within a given set. This method is based on the assumption of circular symmetry around a point maximum (Fisher, 1953) and was described by Ramsay (1967), Cheeney (1983), and Fisher et al. (1987). Each fracture measurement is first converted into a unit direction vector (i.e., pole to the plane). The resultant vector, $\mathbf{R}$, is calculated as the sum of all direction vectors and defines the maximum direction, or mean pole, of the fracture set. Length R is given by:

$$R^2 = \left(\sum_{i=1}^N l_i \right)^2 + \left(\sum_{i=1}^N m_i \right)^2 + \left(\sum_{i=1}^N n_i \right)^2, \quad (1)$$

where l_i , n_i , and m_i are the direction cosines of the individual unit vectors. R is a measure of the dispersion of the data and cannot be greater than N , the number of measurements. In an ideal case, where all measurements are exactly the same, R is equal to N . Increasing discrepancy between R and N indicates increasing dispersion of measurements and greater error in calculating the maximum. The error in the maximum can be quantified in three additional ways: precision, k , is defined as:

$$k = (N-1)/(N-R); \quad (2)$$

spherical variance, v , is defined as:

$$v = (N-R)/N; \quad (3)$$

and a confidence cone, α , can be calculated as:

$$\cos(\alpha_{cl}) = 1 - (N-R)(a-1)/R, \quad (4)$$

where cl is the confidence level, $a = P^{1/(1-N)}$ and $P = (1-cl)$. Commonly, a 95% confidence cone is reported, and P for this case is equal to 0.05. Where a collection of fracture measurements approaches the ideal cluster, all direction vectors equal the mean direction vector, R approaches N , k approaches infinity, v approaches 0, and α_{95} approaches 0° .

FIELD DATA

GPS Data

Three-dimensional spatial data were collected across six pavements (Fig. 3). These pavements were chosen for their locations on the fold in various structural positions (backlimb, hinge, forelimb) and for their noticeable shape differences. Pavement BL1 is in the backlimb; H1 and H2 are in the hinge; FL1, FL2, and FL3 are in

the forelimb. Figure 3 shows the location of the pavements, the collected data points, and preliminary meshes through these points that are color coded for elevation. These help to visualize the dimensions and shape of the pavements. The forelimb pavements are the most limited in area, due to erosion of the steeply dipping strata. The smallest pavement characterized is FL2, which has dimensions of ~6 m long by a few meters wide. The largest pavement is H1, which has dimensions of ~15 m by 10 m.

For data collected with the two-receiver setup and the moving rover (H1 and H2), precision is ~10 cm horizontally and 5 cm vertically. Data for BL1, FL1, FL2, and FL3 were collected with the laser range finder, which adds an additional uncertainty of $\sim \pm 0.1$ m (LaserCraft specification sheet).

Fracture Data

Fracture sets with average strikes of 020° , 045° , 065° , 080° , 135° , and 170° are present within the collected data at the surveyed sites. Many of these sets have been documented in previous Sheep Mountain fracture studies (Fig. 4; Bellahsen et al., 2006a, 2006b); their relative times of formation are determined based on abutting relationships in the backlimb. The 045° set is the hinge-perpendicular set II of Bellahsen et al. (2006a, 2006b) that initiated during early folding in response to fold perpendicular Laramide contraction. The 135° set is the hinge-parallel set III of Bellahsen et al. (2006a) that could have formed at any time during folding. Members of the 135° set are localized in the hinge, but also are found in specific locations in the backlimb and are hypothesized to be related to areas of significant curvature. In some locations, particularly toward the northwest nose of the fold, this fracture set has an average local strike close to 150° . The 080° set apparently formed in response to a rotation in the local most tensile stress direction due to either a change in the regional contraction direction or the influence of a secondary fault developing beneath the backlimb and forming the thumb fold (Fig. 1). The 020° set comprises a minor set documented by Bellahsen et al. (2006a). Based on abutting relationships, 020° fractures are younger than 045° and 080° fractures and older than at least some of the 135° fractures, although most 020° and 135° intersections are crosscutting. The 170° set was identified by Bellahsen et al. (2006a) as a minor set. Although they are formed primarily in the forelimb, these 170° fractures also are found at the backlimb and hinge sites included in this study. Abutting relationships indicate that the 170° set is younger than the 080° set. Based on their preferential formation in specific struc-

tural locations (forelimb, nose), we suggest that this 170° fracture set is folding related.

By assuming that fractures striking 045° and 170° are fold related and determining through the examination of abutting relationships that these two fracture sets are the oldest and youngest studied fracture sets present at the surveyed pavements, we have made the case that all of the fracture sets included in this study formed during folding. The majority of fracture timing relationships were worked out in the backlimb, where abutments are consistent. In other structural positions, however, timing relationships are more uncertain. Opposite abutments exist within the pavements in the hinge and forelimb. We suggest that rather than the formation of each fracture set being confined to specific periods of folding, infilling of previously formed fracture sets (e.g., Bergbauer and Pollard, 2004) occurred throughout folding in the hinge and forelimb as a result of anisotropy within the rock generated by the presence of previously formed fractures.

For this study, the fracture characteristics we focus on are orientation relative to horizontal bedding and intensity. All the fracture sets included in this study are approximately perpendicular to bedding, although there is a fracture set in the backlimb of the anticline that is oblique to bedding. Unfolding observed fracture measurements clarifies which fracture sets in the steeply dipping forelimb are equivalent to sets observed in the more shallowly dipping backlimb and hinge. Observing the orientations of fractures relative to horizontal bedding thus provides the opportunity to study the differences in characteristics of sets developed in different structural positions. In the essentially uniformly dipping beds of the forelimb and the backlimb, the unfolding process does not affect the statistics of the fracture orientations, as all fractures are rotated the same amount. In the hinge, unfolding the hinge-parallel curvature-related fracture set (135°) tightens the clustering. Curvature-related fractures form parallel to the fold hinge and perpendicular to bedding. As mapped in the field, set 135° fractures thus have a larger dispersion in orientation than as considered relative to horizontal bedding. To further investigate the effect of unfolded bedding on fracture set orientation, we tested the 080° set that formed obliquely to the hinge at H2 and found that the spherical variance of the set as measured in the field (0.094) is slightly more than as unfolded (0.091). We suggest that this small difference (0.003) is due to the fact that bedding dip at H2 varies by 5° at most. The surveyed pavements are not expansive enough to generate large bedding dip-related apparent orientation dispersion; the calculated dispersion

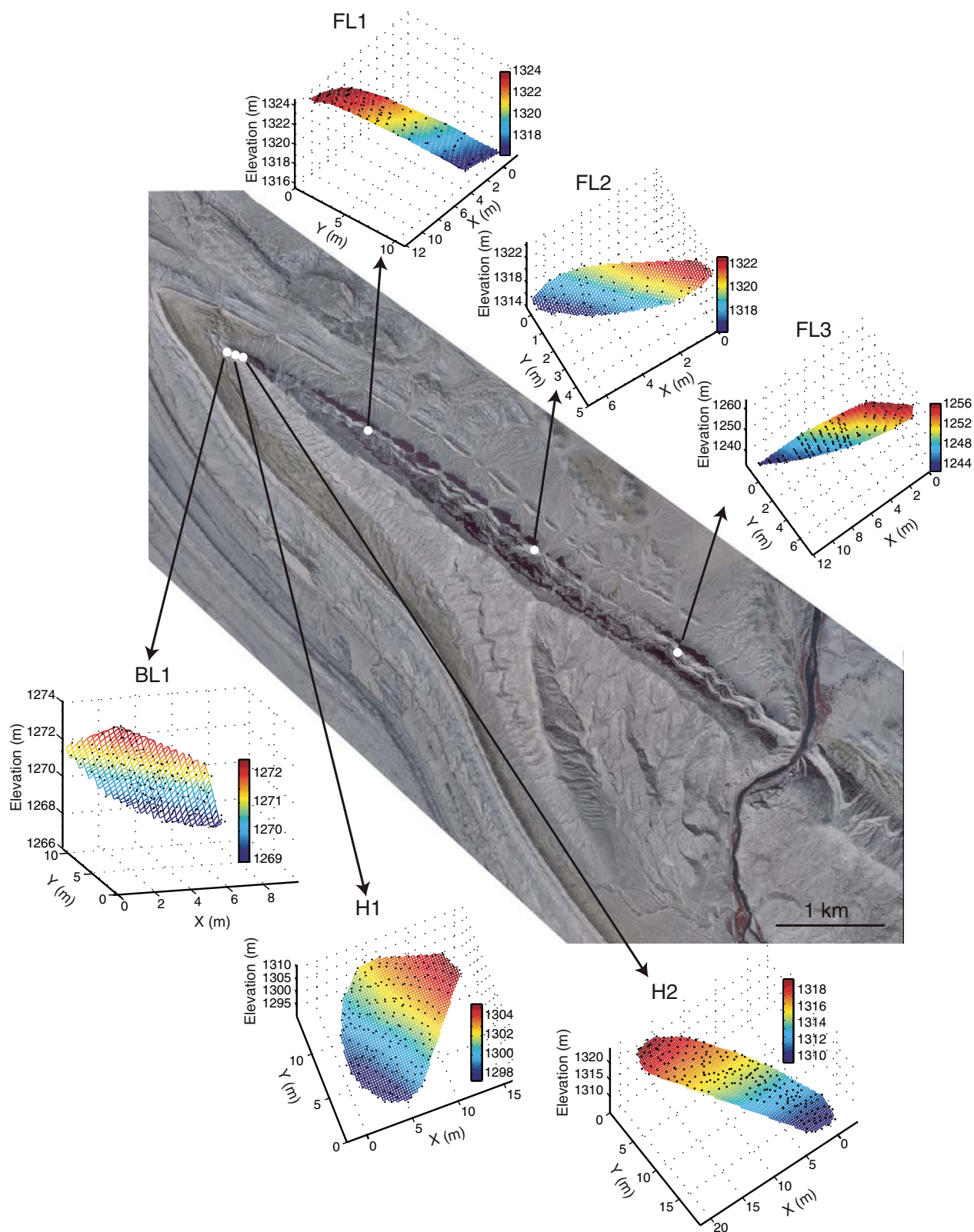


Figure 3. Digital orthophoto quarter quadrangles (DOQQs) of the northwestern part of Sheep Mountain anticline showing the locations of pavements across which global positioning system data were collected and meshes generated from the post-processed data. Meshes are color coded for elevation and provide information on the aerial extent of the mapped pavements. The x axis represents distance across the pavement in the east-west direction. The y axis represents distance across the pavement in the north-south direction. The z axis plots the elevation of the pavement. DOQQs downloaded from <http://wgjac.state.wy.us/>.

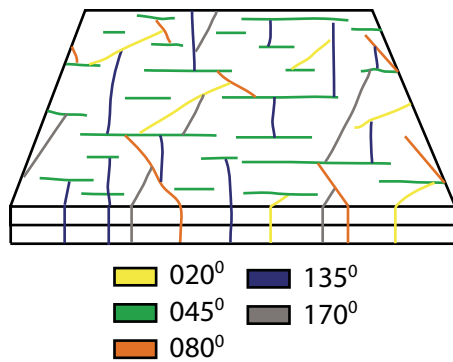


Figure 4. Schematic figure showing abutting relationships among studied fracture sets.

is mostly (97% in the tested case) real. Nonetheless, we unfold fracture measurements before performing Fisher analysis.

Stereonets in Figure 5 show the distribution of poles to fractures for each of the study sites. The data are presented both as observed and unfolded so that bedding is horizontal. Symbols represent fractures of specific sets. Contours within the stereonets indicate where clustering of poles occurs. Rather than being interpreted from stereonets, distinct fracture sets were noted in the field (Fig. 6). Thus, despite a cluster on a stereonet with one maximum that seems to represent a single fracture set, in some cases two sets within the cluster have been recorded (e.g., sets 140° and 159° at BL1; Figs. 5A and 6A).

Results of a Fisher analysis for each of the fracture sets are listed in Table 1. We consider values of spherical variance to compare the dispersion of specific sets in different structural locations. The 020° set exists at one site in each position: BL1 in the backlimb, H2 in the hinge, and FL1 in the forelimb. Spherical variance is lowest at FL1 and highest at H2. The 045° set has a lower spherical variance at BL1 in the backlimb than at H1 in the hinge. The 080° set has comparable spherical variance values at backlimb and forelimb sites BL1, FL1, and FL2; it has higher values at hinge sites H1 and H2. Although no fractures of the set with an average strike of 135° have been recorded in forelimb sites, the spherical variance of this set at backlimb site BL1 is much less than that at either hinge site. The spherical variance for the 170° set at BL1 is intermediate to the values at forelimb sites FL1 and FL3. All three of these values are less than the spherical variance of the 170° set at hinge sites H1 and H2.

Fracture intensities for the six distinct fracture sets are represented by bar graphs in Figure 7. Average thickness of the bedding at each pavement is also documented in Figure 7. We do not normalize the intensities to bed thickness

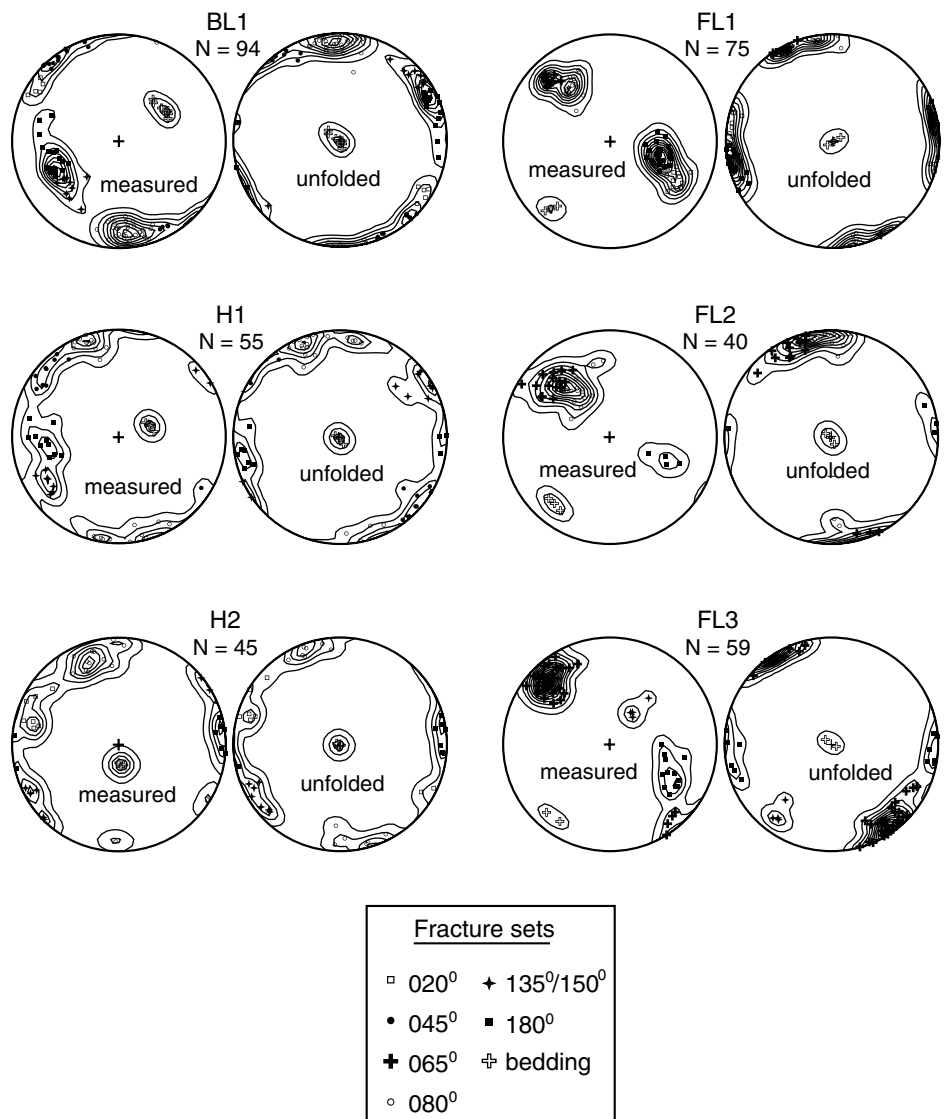


Figure 5. Fracture orientation data collected at each pavement. Lower hemisphere stereonets show the orientations of poles to fractures as observed in the field (measured) and relative to horizontal bedding (unfolded). Specific fracture sets were defined in the field based on orientation, mode of deformation, and abutting relations. The average strike direction of each fracture set is in the figure legend, next to the corresponding symbol that has been used to plot the poles for that set. Contours are plotted at an interval of two and highlight the clustering of poles.

because all thicknesses are similar. Values plotted represent the number of fractures per meter measured along a line that is perpendicular to the fracture strike. Highest intensities occur in the Phosphoria pavements that are in the hinge: H1 and H2. Lowest intensities occur in the backlimb Phosphoria pavement BL1. Forelimb intensities are intermediary. The intensity of the fracture set with an average strike of 080° is approximately consistent between the hinge and the forelimb, with values at H1 and FL2 similar and values at H2 and FL1 similar. The intensity

of the same 080° set at site BL1 in the backlimb is much lower. In the backlimb at BL1 and the forelimb at FL3, the intensities of the 135° set are similar. The 135° set is more intense at both hinge locations. Where the 170° set exists in the hinge at H2, it is three times more intense than at FL1. The other two forelimb sites and the backlimb site BL1 have intensities of the 170° set that are less than that at FL1. In the backlimb, no fracture set has an intensity more than two fractures per meter. In the hinge, all fracture sets have intensities greater than six

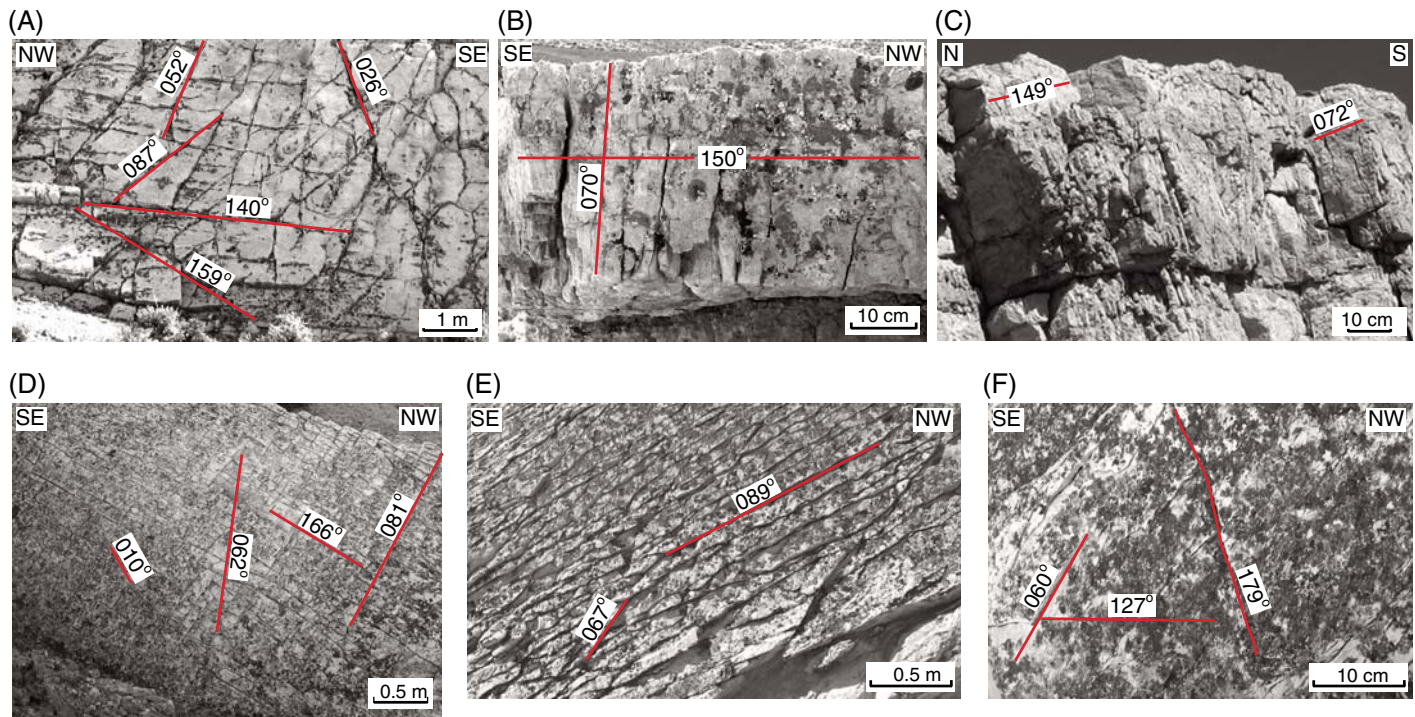


Figure 6. Fracture sets present at pavements. (A) BL1. (B) H1; sets with average strikes of 040° and 174° also are present at this site. (C) H2; sets with average strikes of 023° and 172° also are present at this site. (D) FL1. (E) FL2; a minor fracture set striking 180° also is present at this site. (F) FL3. Clusters in Figure 5 were broken apart based on these field observations. B and C are cross-sectional views of fractures in relatively flat bedding planes. A, D, E, and F are photographs of bedding surfaces.

fractures per meter. In the forelimb, the intensity of fracturing varies from set to set, and even within the same set. The 065° set, for example, varies from less than one fracture per meter at FL1 to approximately two fractures per meter at FL2 to approximately six fractures per meter at FL3. At each of the three forelimb sites, a different fracture set is dominant.

Curvature Analysis

We begin the curvature analysis by removing an average plane through the data. This process effectively removes the dip of the data set, allowing one to better compare the relative elevations of the collected data points (Fig. 8). The relative elevations color coded in Figures 8A, 8D, 8E, and 8F are very noisy, indicating that there are no consistent elevation trends within the data. Conversely, red swaths in Figures 8B and 8C represent maxima, and suggest the presence of

TABLE 1. FRACTURE SET STATISTICS

Pavement		Mean plane ($^\circ$)	N	R	v	k	95% ($^\circ$)
BL1	□	026/87	7	6.85	0.141	38.86	9.8
	●	052/89	12	11.83	0.076	65.58	22.8
	○	087/83	35	32.56	0.042	13.93	6.7
	✦	140/83	10	9.88	0.095	63.15	6.1
	■	159/83	19	18.75	0.042	72.87	4.0
H1	●	040/89	10	7.95	0.117	4.39	26.1
	○	070/83	15	9.05	0.354	2.35	32.5
	✦	150/84	10	5.22	0.348	2.11	47.9
	■	174/84	10	9.78	0.087	41.02	7.6
H2	□	023/79	8	5.94	0.152	3.4	35.5
	○	072/86	13	10.91	0.091	5.73	19.0
	✦	149/75	7	2.96	0.507	1.48	83.4
	■	172/83	8	3.99	0.431	1.74	62.5
FL1	□	010/88	14	13.87	0.067	102.2	4.0
	+	062/88	12	11.96	0.087	268.7	2.7
	∪	081/89	17	16.82	0.051	89.09	3.8
	■	166/87	29	28.66	0.023	81.45	3.0
FL2	+	067/83	18	17.64	0.038	47.68	5.1
	∪	089/86	14	13.63	0.049	35.59	6.8
FL3	+	060/87	41	32.81	0.18	4.89	11.3
	■	179/89	12	11.73	0.066	40.03	6.9

Note: Fisher statistics as calculated for each of the fracture sets measured at individual pavements. The mean fracture plane, number of measurements (N), magnitude of the resultant vector (R), spherical variance (v), precision (k), and 95% confidence cone (95%) are tabulated. Symbols reference each fracture set to those defined in Figures 5, 7, and 12.

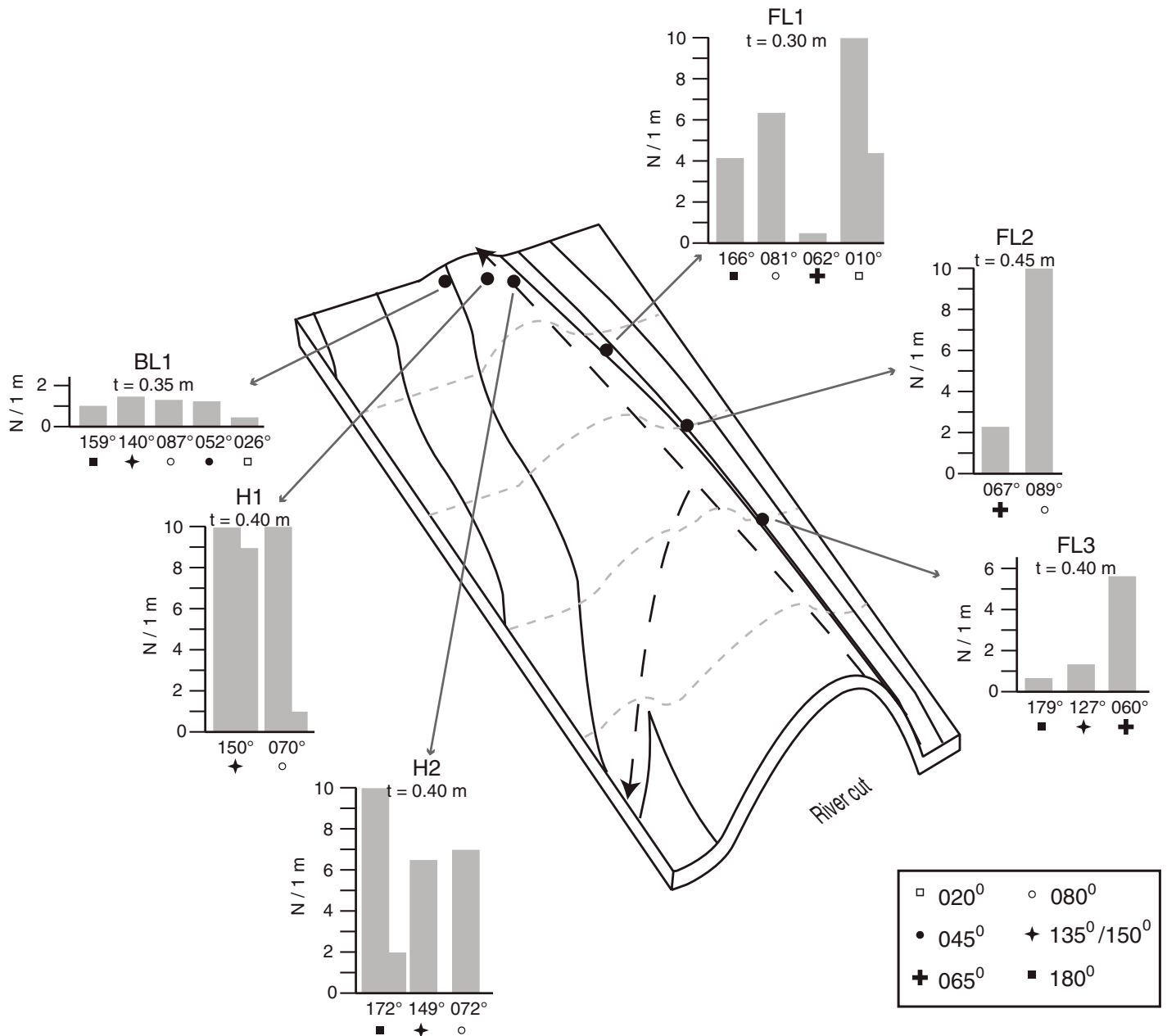


Figure 7. Three-dimensional sketch of Sheep Mountain with bar graphs showing fracture intensities of the major fracture sets at each study site. Intensities have been normalized to show the number of fractures per meter. To facilitate comparison of fracture intensities among sites, symbols are plotted beneath the bar graph to indicate which sets have been determined to be equivalent (based on orientation, mode of deformation, abutting relations) at different sites. An average bed thickness (t) for each pavement is reported.

anticlinal curvatures. Application of curvature analysis to the data sets in their current state would highlight the small-scale undulations visible in Figure 8. These undulations are most likely a result of slight spatial differences in erosion or the error included in the data collection method, and do not reflect the natural shape of the bedding surface. To remove these artifacts from the data sets and thus capture folding on the appropriate scale, we followed the spectral analysis technique

of Bergbauer and Pollard (2003). Application of a smoothing factor that removes oscillations with wavelengths <13 m discards the high-frequency content of the data sets while preserving surface trends. This minimum wavelength was determined through a sensitivity analysis in which the removed data were examined. For filtered wavelengths ≤ 13 m, the removed data represented incoherent noise. When data of wavelengths >13 m were filtered, the removed data

began to show trends, indicating that data relating to the shape of the surfaces had been removed. The smoothed surfaces are shown in Figure 9. The lack of contours in Figures 9A, 9D, 9E, and 9F indicates that the surface elevations vary by <0.1 m and are thus approximately planar.

The maximum and minimum principal curvatures, κ_1 and κ_2 , were calculated for the area surrounding each grid node of the smoothed surfaces. The extreme values of κ_1 and κ_2 for

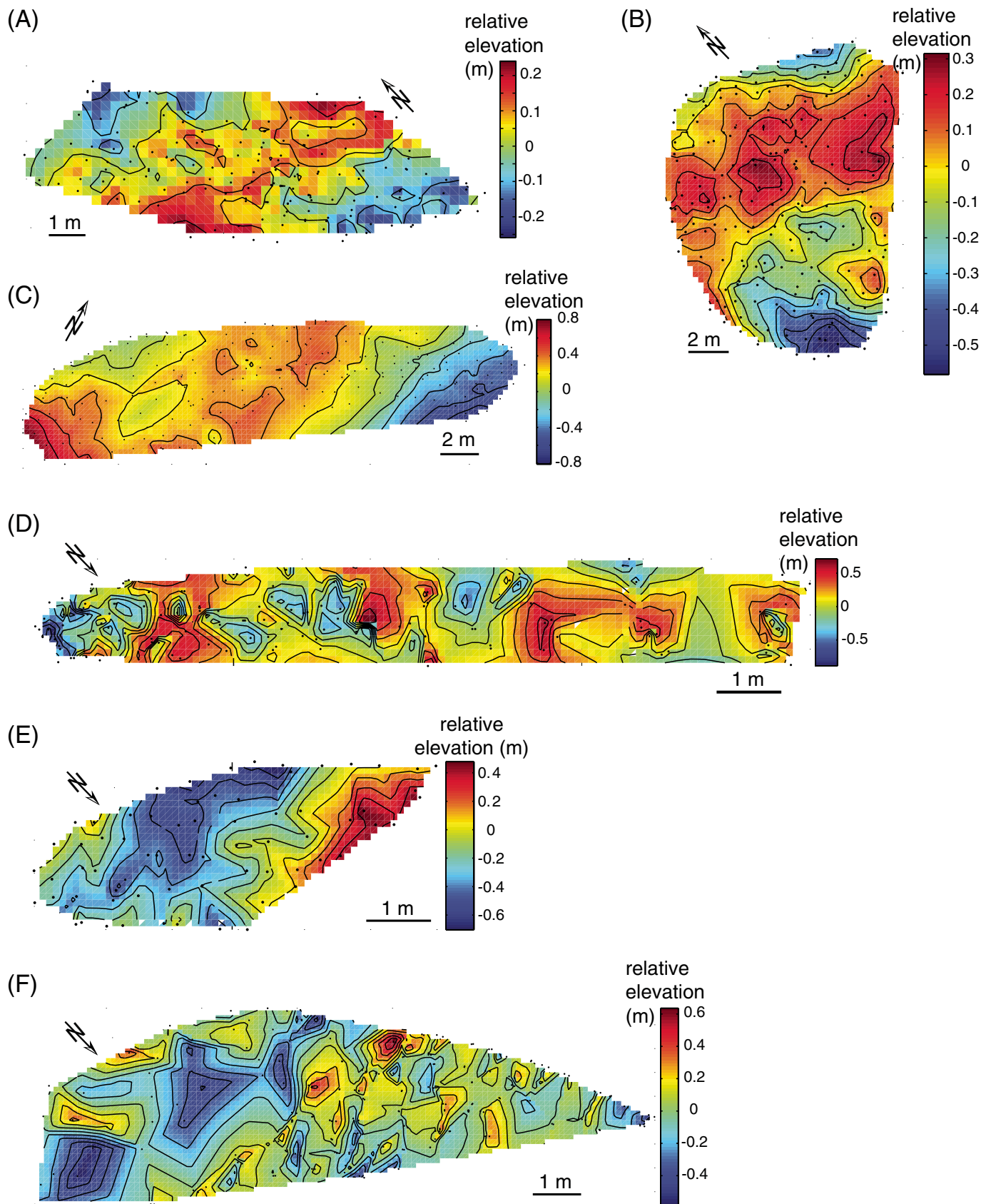


Figure 8. Relative elevations across surfaces constructed from unfiltered global positioning system data at pavements (A) BL1, (B) H1, (C) H2, (D) FL1, (E) FL2, and (F) FL3.

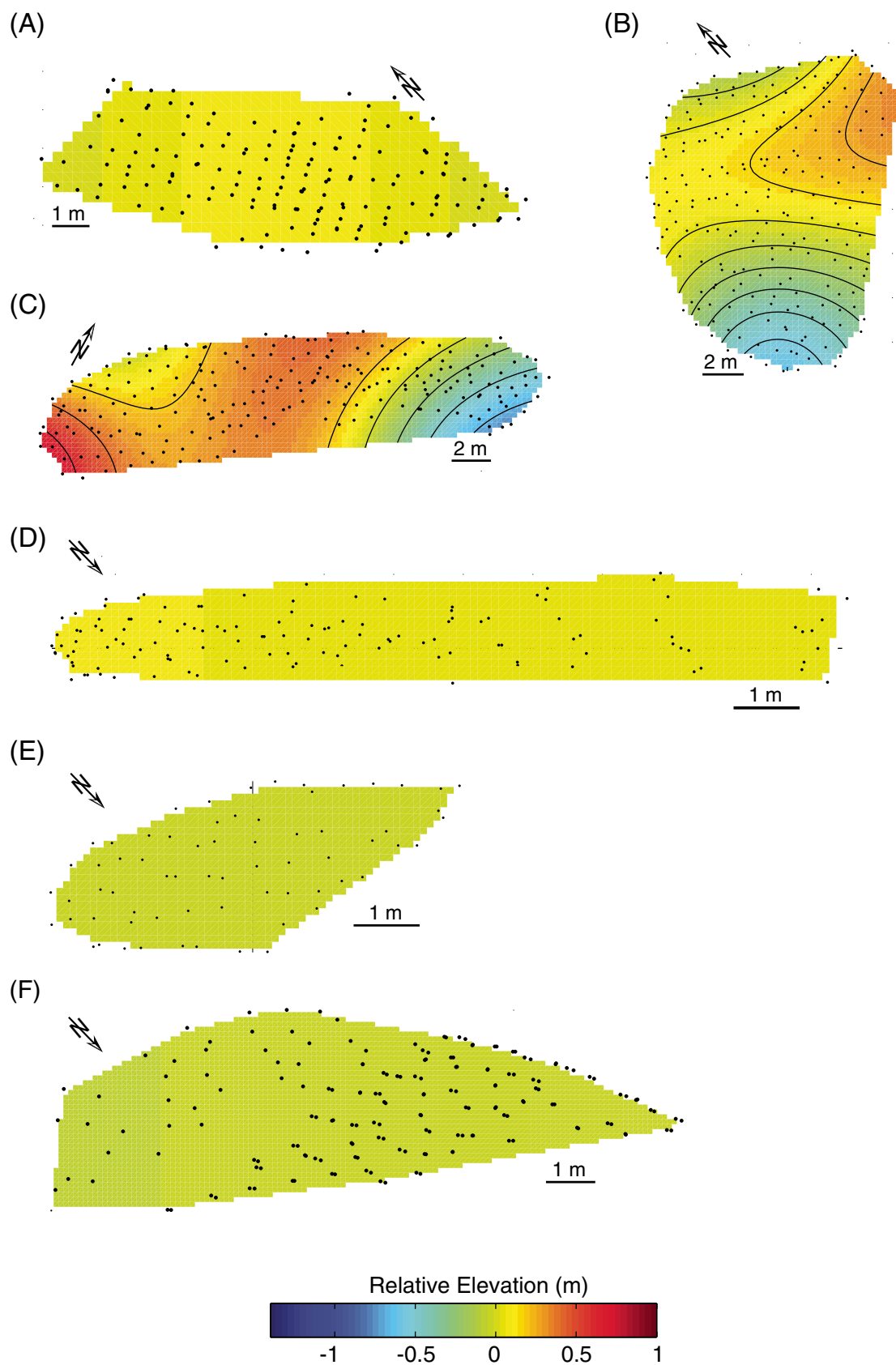


Figure 9. Smoothed relative elevations across surfaces constructed from filtered global positioning system data collected at pavements (A) BL1, (B) H1, (C) H2, (D) FL1, (E) FL2, and (F) FL3. Contour interval is 0.1 m.

TABLE 2. MAXIMUM AND MINIMUM NORMAL CURVATURE

Pavement	κ_1 (m^{-1})	κ_2 (m^{-1})
BL1	0.0086	-0.0082
H1	0.0399	-0.0442
H2	0.0466	-0.0340
FL1	0.0010	-0.0010
FL2	0.0004	-0.0003
FL3	0.0034	-0.0038

Note: Maximum principal normal curvature (κ_1) and minimum principal normal curvature (κ_2) calculated across each pavement.

each surface are listed in Table 2, which indicates that BL1, FL1, FL2, and FL3 have very small curvatures, with extreme values ranging from 0.00034 m^{-1} to 0.001 m^{-1} . The spectral analysis filtering algorithm removed all of the noise from the original surfaces (Fig. 8), producing smoothed surfaces (Fig. 9) with curvatures that are negligible, of magnitude less than the value of 0.005 m^{-1} that we designate as the curvature threshold. When we apply the threshold value, H1 and H2 are the only two pavements that have significant curvatures. The values and directions of κ_1 and κ_2 across these surfaces are plotted in Figure 10 with a normalized color bar so that magnitudes of curvature can be compared between the data sets. Gaussian curvature also was computed and evaluated, but did not add further insight to the study.

DISCUSSION

Curvature Analyses

In the maximum and minimum principal curvature plots (Fig. 10), warm colors represent anticlinal folding and cool colors represent synclinal folding. H1 and H2 are very near to the hinge line of the fold (Fig. 5). To the naked eye, these surfaces appear to have distinct curvatures (Fig. 11A). Analysis confirms the existence of this curvature (Fig. 10). As expected, H2, which is closer to the hinge line, has higher magnitudes of curvature than H1. The curvature analysis for H1 highlights that the surface has nonzero magnitudes of normal curvature in the two principal directions, one parallel to the hinge of the fold and one perpendicular to the hinge of the fold. This phenomenon is noticeable in field photos (Fig. 11A). Localized flexures of bedding surfaces also are found in the nose. The curvature analysis for H2 reflects one instance of this localized flexure. In Figure 10C, the left edge of the data set represents the area closest to the hinge. A traverse from left to right across this plot approx-

imates the dip azimuth. In the curvature plot, the existence of anticlinal curvature across the majority of the pavement is apparent, but an area of synclinal curvature exists at extreme downdip locations. This synclinal curvature also is noticeable in field photos (Fig. 11B).

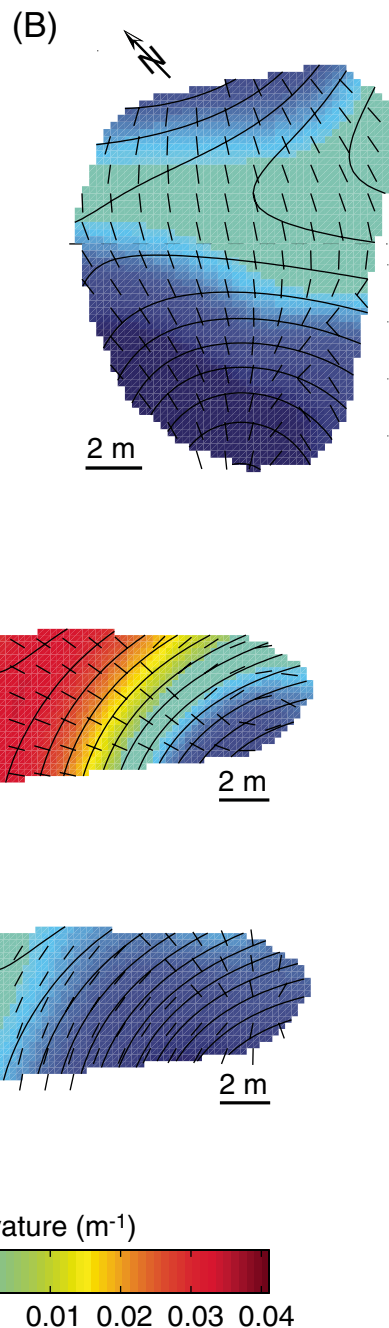


Figure 10. Plots of maximum (κ_1) and minimum (κ_2) normal curvature values across surfaces for which magnitudes are greater than a threshold value of 0.005 m^{-1} . The corresponding directions of κ_1 and κ_2 are represented by black tick marks. (A) κ_1 values and directions across H1. (B) κ_2 values and directions across H1. (C) κ_1 values and directions across H2. (D) κ_2 values and directions across H2.

Relating Curvature Analysis to Fracture Measurements

The most noticeable correlation between curvature and fracturing relates to fracture orientation. In Figure 12, we plot spherical variance

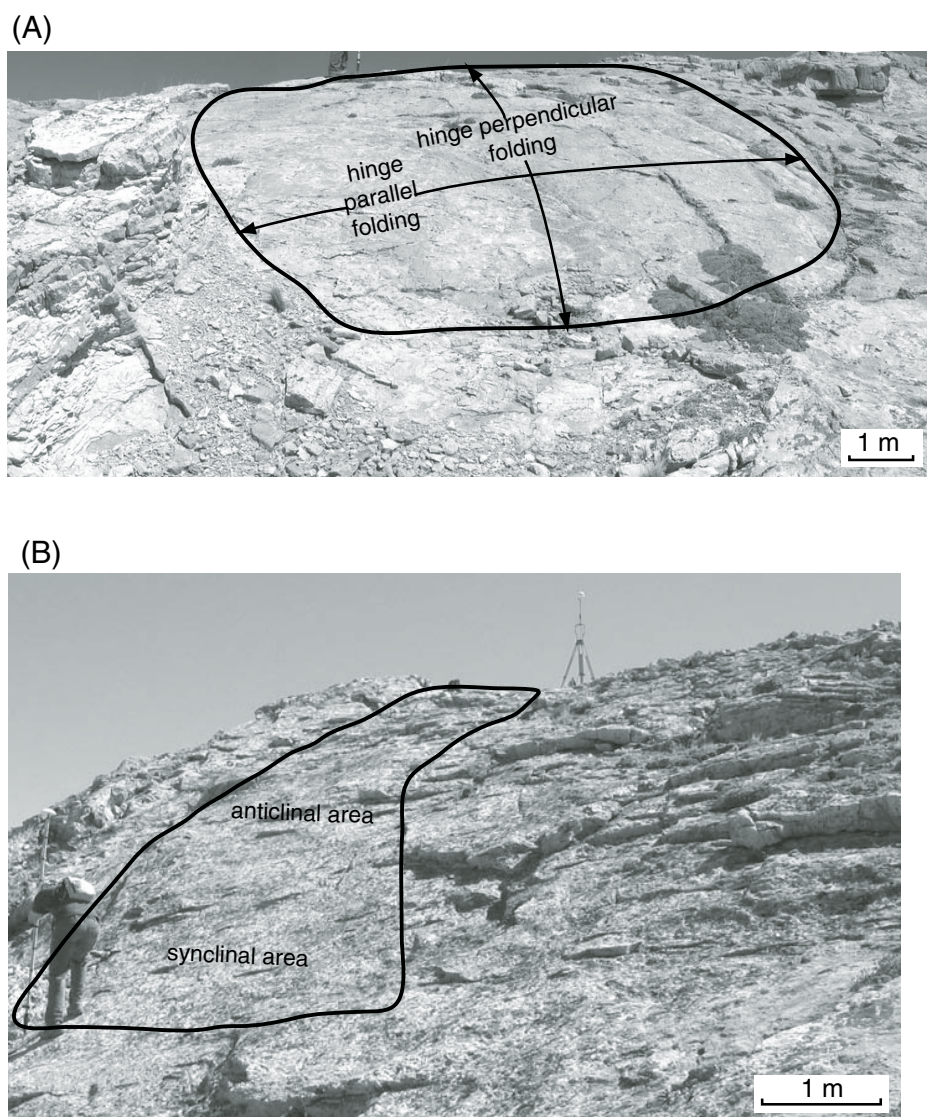


Figure 11. Photographs of the pavements at (A) H1 and (B) H2. The black line outlines the approximate area of surveying. In A, two directions of curvature are apparent, one parallel to the hinge and one perpendicular to the hinge. In B, both anticlinal and synclinal curvature are apparent in a direction parallel to the hinge. In the photo, shading highlights the faces of fractures striking $\sim 080^\circ$.

for the fracture sets mapped in this study. Large spherical variances occur only at sites H1 and H2, the two pavements of notable curvature. Thus, in pavements that have higher curvature values, the clustering of distinct fracture sets is less pronounced than in pavements of lower curvature values.

We look at minimum normal curvature (κ_2) trajectories to determine if fracturing was enhanced by folding in the direction perpendicular to the maximum curvature. At pavement H1, κ_1 is parallel to the hinge. Thus, curvature-related fractures would be expected to form in the dip direction (Fig. 10B), which ranges from

065° to 080° . Both the intensity (Fig. 7) and the spherical variance (Table 1) of the fracture set of average strike 070° are large. However, two directions of curvature are present at this pavement (Fig. 11A). Fracturing may thus be expected in the direction perpendicular to this set, formed at a strike of $\sim 160^\circ$. The spherical variance and intensities of a set striking 150° at H1 are also large. At H2, κ_2 is parallel to the hinge (Fig. 10D). H2 is at a location in the hinge where the plunging northwestern nose controls the local strike direction, which varies between 077° and 100° . Curvature-related fractures would thus be expected to form parallel to

this direction. Although the spherical variance and intensity of the 072° set is great compared to sets at pavements BL1, FL1, FL2, and FL3, the values of spherical variance for the 149° and 172° sets at H2 are greater. The intensity of the 149° set is slightly less than that of the 072° set, and the 172° set is much more intense. Fracture statistics at H1 and H2 suggest that, if fracturing in the hinge is due primarily to curvature, sets in addition to the fracture set that forms in the minimum curvature direction due to outer arc extension of a flexed bedding surface (Timoshenko, 1934) are affected by folding.

Fracture intensities do not correlate directly with curvature. Although the highest intensity of fracturing and the highest values of curvature are found at H1 and H2 in the hinge (Fig. 7), high intensity of fracturing also has been recorded in the forelimb at FL1, FL2, and FL3, where the lowest curvature magnitudes exist. Fracturing in the forelimb is not related to present-day curvature. It is most likely not related to paleo-curvatures (i.e., migration of forelimb beds through the hinge) either, as characteristics of fracturing in the hinge and forelimb are very different. Spherical variances of all fracture sets in the forelimb are small (Table 1), and the hinge-parallel fracture set striking 135° is sparse; whereas in the hinge, large spherical variances are found and the 135° fracture set is intensely formed. We suggest that the fractures in the forelimb formed due to some mechanism other than bedding flexure. Noting that the fracture sets developed in the forelimb, as well as the most intensely formed set, vary from site to site, we suggest that fracturing has been affected by some local mechanism of deformation. Curvature is only one mechanism for fracture development. Other mechanisms such as stretching, faulting, bedding-plane slip, the influence of preexisting fractures, and tectonic loading may contribute to fracture development and should be evaluated. At structures similar to Sheep Mountain anticline, a complete understanding of the heterogeneity in fracture intensities requires analysis beyond curvature calculations.

CONCLUSIONS

GPS collection and post-processing methods have allowed us to acquire very precise three-dimensional surface data for pavements within the Phosphoria Formation at Sheep Mountain. Curvature analyses of, and fracture measurements across, these surfaces indicate that greater curvature correlates with greater variance in fracture orientation. Fracture intensities do not correlate directly with curvature, indicating that additional mechanisms of fracture development must be evaluated.

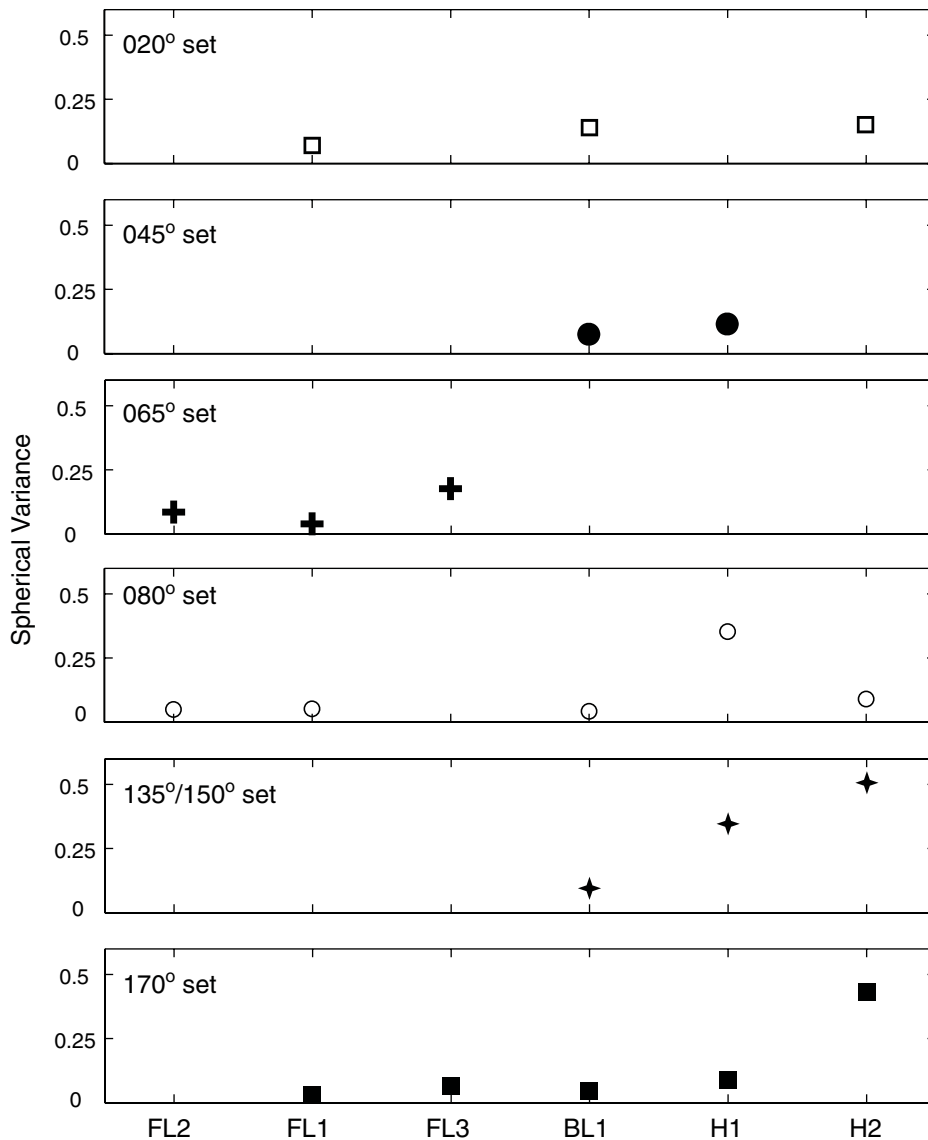


Figure 12. Spherical variance for the fracture sets mapped at study sites. Pavements are plotted on the x axis in the order of increasing curvature magnitudes. Symbols represent specific fracture sets as defined in Figures 5 and 7.

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